

Snort is not the negative of Col: a counterexample to Conjecture 00000008887

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Take the graph K_2 , with two vertices joined by one edge. In Snort, the first player wins. In Col, the second player wins. Negating Col exchanges the two players' options, so the second player still wins. Therefore

$$\text{Snort}(K_2) \neq -\text{Col}(K_2).$$

The graph has a nonidentity automorphism exchanging its two vertices, so it is not covered by the source's proposed exception for asymmetric graphs.

Rules and complete play analysis. Both games use normal play: the player unable to move loses. Each player places their own color on an empty vertex. In Snort, adjacent vertices may not have opposite colors; in Col, adjacent vertices may not have the same color. These are the standard rules; see Definition 1.4 in Games of No Chance 6.

In Snort on K_2 , the first player colors either vertex. The remaining vertex is adjacent to the opponent's color, so the next player has no legal move. This is a first-player win, irrespective of who starts. In Col, after the first move the other player can color the other vertex: it is adjacent to the opposite color, which is permitted. No empty vertex remains, so the starting player loses. These cases exhaust every possible play from the empty board. Equivalently, the game forms are

$$\text{Snort}(K_2) = \{1 \mid -1\}, \quad \text{Col}(K_2) = \{-1 \mid 1\} = 0.$$

The outcome separation alone suffices; the proof does not require the numeric simplification of the second form.

Formal verification. The Lean file defines a finite normal-play partisan game by its complete state list, all Left and Right moves, and a strictly decreasing integer height. Its recursive winning predicate examines every legal move and declares a terminal player losing. A separate finite-depth evaluator is proved equal to this well-founded predicate whenever its depth exceeds the position height; the computation is therefore not an unverified truncation of play.

The concrete board is a pair of colors in $\{0, 1, 2\}$, giving all nine colorings. The file proves that every function from the two vertices to the three colors is represented. Every move is proved to be exactly a legal placement at one of the two vertices, and every such move strictly decreases the number of empty vertices. The adjacency relation, absence of loops, and nonidentity vertex-exchange automorphism are also checked.

Negation interchanges Left and Right moves at every state. Disjunctive addition is defined for arbitrary finite games. The predicate `EqualValue` expresses equality of outcomes after adding every finite normal-play context, for either starting player. This is the standard observational definition of equality of short-game values. The theorem `conjecture8887_counterexample` disproves it for Snort and negative Col by specializing to the zero-game context, where their outcomes are respectively winning and losing. Thus the formal conclusion compares actual games, including the negation operation, rather than merely comparing names or assigned numerical labels.

The proof uses Lean 4.19.0 with `Std` only. All finite computations are checked by the kernel; there are no admitted lemmas, native evaluation proofs, or custom axioms.